



• HUMAN-CENTERED INDUSTRIAL AI SOLUTION

PS 26099 — AI-Driven Standardization and Harmonization of Material Codes Across CPSEs

Theme: Smart Automation • Building a Unified National Material Master for CPSEs

1. THE PROBLEM Fragmented CPSE Material Data

CPSEs across Oil & Gas, Power, Steel, Mining & Heavy Engineering maintain large material master databases with inconsistent codes, names, descriptions, specifications, units of measurement, classifications, and ERP/SAP systems — leading to disorganized, siloed material data.

Example: The Same Material Recorded Differently Across CPSEs

CPSE A MAT-10452 "SS PIPE 2 IN SCH 40"	CPSE B M-77821 "STAINLESS STEEL PIPE, 50mm, SCH40"	CPSE C RM-2941 "SS SEAMLESS PIPE DN50 SCH 40"
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Resulting Consequences

- Duplicate Material Masters
- Inconsistent Data & Poor Data Quality
- Difficult Material Discovery & Search
- Fragmented Procurement Visibility & Higher Inventory Costs

2. THE INTELLIGENCE Our AI-Powered Solution

National Unified Material Master

An AI-powered platform for discovering, standardizing, and harmonizing equivalent materials across CPSEs. AI recommends • Human expert validates • Full traceability maintained.



HUMAN-IN-THE-LOOP GOVERNANCE

Controlled Master Governance

- AI provides recommendations with confidence score, never auto-commits changes
- Domain expert reviews, approves, or edits before updating master data
- Full audit trail & version history for compliance and governance

• Possible equivalent materials detected
Reason: Same material grade (SS 304) + dimensions (DN50 / 2" / 50mm) + pressure class (SCH 40)
Match confidence: 96%

[Approve Mapping]

[Review Details]

[Reject]



Code Mapping & Traceability

Zero ERP Disruption

OLD CPSE CODES (LEGACY)

CPSE A: MAT-10452	-
CPSE B: M-77821	-
CPSE C: RM-2941	-



UNIFIED NATIONAL CODE

NUMM-SS-PIP-050-S40

Standardized: SS Pipe | DN50 | SCH40 | Seamless | Traceable to all source codes

Live Platform Dashboard (Demo)

*Illustrative Demo Values

Materials Analysed 128,450	Potential Duplicates Found 9,342
Equivalent Groups Identified 2,117	Pending Expert Validation 421

Intended Benefits for CPSE Ecosystem

Value Proposition

- ✓ **Cost Optimization**
Reduce duplicate inventory & holding costs through consolidation
- ✓ **Operational Efficiency**
Faster material search & procurement decisions
- ✓ **Supply Chain Visibility**
Unified view across CPSEs for strategic sourcing
- ✓ **Compliance & Standardization**
Single source of truth for reporting & audit

From fragmented material records →
to one intelligent material identity.

TECH STACK:

NLP Semantic Search Embeddings Entity Matching ML

Technical Attribute Extraction • Classification • Confidence Scoring • Human-in-the-Loop • SAP/ERP Integration